



Modelling in Mechanics · Mark Scheme

Mechanics Chapter 1: Modelling in Mechanics. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Modelling assumptions

- 1 Light: the string has no mass *B1*. Inextensible: the string does not stretch *B1*.
- 2 E.g. the size of the ball, so air resistance/spin/rotation is ignored *B1 B1*.
- 3 (a) There is no friction between the book and the table *B1*. (b) A real table is rough, so friction acts *B1*; the book would slow down and stop, rather than move at constant speed *B1*.
- 4 Light: the rope's own weight can be ignored *B1*; inextensible: the rope stays the same length, so the tension acts along it and the crate moves with the end of the rope *B1*; particle: the crate's size and rotation are ignored, and all forces act at one point *B1*.
- 5(a) E.g. no air resistance; the stone is a particle; g is constant *B1 B1*.
- 5(b) Air resistance would reduce the acceleration, so the time would be longer *B1*.
- 5(c) Include air resistance *B1*; use a more accurate value of g for the location / consider the stone's size and shape *B1*.
- 5(d) A leaf is light with a large surface area *B1*, so air resistance is significant and cannot be ignored (it also flutters and rotates) *B1*.

Section B – Quantities and units

- 6 Velocity: vector, time: scalar, force: vector, distance: scalar *B2* (*B1* for 2 or 3 correct).
- 7 (a) $72 \times 1000 \div 3600 = 20 \text{ ms}^{-1}$ *B1*; (b) 0.45 kg *B1*; (c) 150 s *B1*.
- 8 (a) $5 \times 9.8 = 49 \text{ N}$ *B1*; (b) $147 \div 9.8 = 15 \text{ kg}$ *B1*; (c) $5 \times 1.6 = 8 \text{ N}$ *B1*.
- 9 (a) Distance 7 km *B1*; displacement $\sqrt{3^2 + 4^2} = 5 \text{ km}$ *B1*. (b) Average speed $7 \div 1.25 = 5.6 \text{ km h}^{-1}$ *B1*; average velocity $5 \div 1.25 = 4 \text{ km h}^{-1}$ *B1*.
- 10(a) Distance 24 m *B1*; displacement 0 *B1*.
- 10(b) Average speed $24 \div 3 = 8 \text{ ms}^{-1}$ *B1*; average velocity 0 *B1*.
- 10(c) $90 \times 1000 \div 3600 = 25 \text{ ms}^{-1}$ *B1*; $W = 1200 \times 9.8 = 11760 \text{ N}$ *B1*.
- 10(d) Weight is a force with a direction (vertically downwards); mass has size only *B1*.

Roar-level challenge

- 11 (a) Distance $= 2\pi(50) = 314 \text{ m}$, average speed $\frac{314.2}{62.8} = 5.00 \text{ ms}^{-1}$ *B1*; displacement 0 , so average velocity 0 *B1*. (b) Displacement $= 100 \text{ m}$ (the diameter) *B1*; average velocity $\frac{100}{31.4} = 3.18 \text{ ms}^{-1}$ *B1*.